

Activity

Group members:

Grad applications data

Using the grad applications data, you fit a model with GPA as the explanatory variables. Here is the regression output:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.3576	1.0353	-4.209	2.57e-05	***
gpa	1.0511	0.2989	3.517	0.000437	***

And here is the variance-covariance matrix for the fitted model:

	(Intercept)	gpa
(Intercept)	1.0718814	-0.3076994
gpa	-0.3076994	0.0893229

Questions

1. Calculate the estimated log odds of admission for a student with GPA 3.8.

Fact: For two variables U, V and constants a, b :

$$Var(aU + bV) = a^2Var(U) + b^2Var(V) + 2abCov(U, V)$$

For a student with GPA 3.8, the estimated log odds is

$$\widehat{\log \text{ odds}} = \hat{\beta}_0 + \hat{\beta}_1(3.8)$$

2. Using the fact above, and the variance-covariance matrix from the R output, calculate $Var(\widehat{\log \text{ odds}})$ when $GPA = 3.8$.

3. From your answer in question 2, calculate $SE(\widehat{\log \text{ odds}})$.

A 95% CI for the *true* log odds of admission for students with GPA 3.8 is

$$\widehat{\log \text{ odds}} \pm 1.96SE(\widehat{\log \text{ odds}})$$

4. Using your answers to questions 1 and 3, calculate a 95% CI for the true log odds of admission for students with GPA 3.8.

5. Finally, transform your confidence interval from question 4 into a 95% CI for the true *probability* of admission for students with GPA 3.8.